

PAPER_962: Resonant Gravity Triadic Mode

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** triadic_solutions_next.py (ResonantGravityTriadic) **Calculator:** ResonantGravityTriadicCalc (CP4 #546) **CVW:** v2.0.0 compliant

Abstract

The Resonant Gravity Triadic mode uses the 1.25 THz phonon linewidth Γ to tune neutron-drop and buoyancy reversal. When $\Phi(\omega_{\text{SCm}}, \Gamma) > \Phi_{\text{crit}}$, the neutron drip-line shifts, controlling NS merger dynamics.

1. Resonant Phonon Occupation

$$\Phi(\omega, \Gamma) = \Phi_0 \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}([SSq])$$

2. Neutron-Drop Threshold

$$\Phi(\omega_{\text{SCm}}, \Gamma) > \Phi_{\text{crit}} \implies \text{neutron-drop triggered}$$

3. Buoyancy Reversal

$$t_{\text{rev}} = \frac{\pi}{2\Gamma}$$

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)